Assignment - I
Cloud Computing

(a) What are the Goals and Benefits of cloud computing?

The common benefits associated with adopting cloud computing are:

1. Reduced investments and proportional costs
2. Increased scalability (elasticity)
3. Increased availability and reliability

1. Reduced investments and proportional costs

Reduction or elimination of upfront capital expenditures - hardware, software and ownership costs

Proportional costs: cloud measured usage allows measured operational expenditures (directly related to business performance) to replace anticipated capital expenditures.

Cost reduction allows capital to be directed to the core business of the company.

Benefits: pay-as-you-go on-demand access, unlimited computing resources, add or remove IT resources at a fine-grain level, applications are not locked into devices or locations can be easily moved.

2. Increased Scalability (Elasticity)

Empower cloud consumers to scale cloud resources to accommodate workload fluctuations and peaks.

Meets unpredictable usage demands and avoid potential loss of business.

3. Increased Availability and Reliability

Tangible business benefits: service outages result in loss revenue

Service outages:

Gmail: Feb 2011 – 120,000 of 150m users without email over the weekend

PayPal: Nov 2010 – offline for 4.5 hours (network equipment failure)

Facebook: Sep 2010 – offline for 2.5 hours (database overload)

Microsoft Azure Cloud: Mar 2008 – down for 22 hours

Dropbox: Feb, 2013 – down for 48 hours (200m users)

Availability: Gmail [99.984% (2010), 99.983% (2012)]

99.984% = downtime per year

(b) What are the most important cloud service models of cloud computing?

The most important cloud service models of cloud computing are

1. Software-as-a-Service (SaaS)
2. Platform-as-a-Service (PaaS)
3. Infrastructure-as-a-Service (IaaS)

1. Software-as-a-Service (SaaS)

Definition:

Use provider's applications over a network, e.g., Salesforce.com, Google apps.

Description:

SaaS delivery model is typically used to make a reusable cloud service widely available (often commercially) to a range of cloud consumers.

The cloud service consumer is given access to the cloud service contract, but not to any underlying IT resources or implementation details.

2. Platform-as-a-Service (PaaS)

Definition:

Deploy customer-related applications to a cloud, e.g., Google's app engine, Amazon AWS, Microsoft Windows Azure.

Description:

A pre-defined "ready-to-use" environment typically comprised of already deployed and configured IT resources.

Common reasons a cloud consumer would use and invest in a PaaS environment:

- To extend on-premise environments into the cloud for scalability and economic purposes.
- To entirely substitute an on-premise environment by using the ready-made environment.
- To become a cloud provider and deploys its own cloud services to be made available to other external cloud consumers.

PaaS products are available with different development stacks. For example, Google App Engine offers a Java and Python-based environment.

3. **Infrastructure-as-a-Service (IaaS)**

Definition:

Rent processing, storage, network capacity and other fundamental computing resources, e.g., Amazon EC2, Savvis Symphony.

Description:

A cloud consumer is using a virtual server within an IaaS environment.

Cloud consumers are provided with a range of contractual guarantees by the cloud provider, pertaining to characteristics such as processor capacity, memory, local storage, performance, and availability.

(c)What are Common characteristics of Virtualization technology in cloud computing?

Virtualization

Virtualization is the process of converting a physical IT resource (like a server or storage device) into a virtual IT resource. It primarily focuses on the creation and deployment of virtual servers using server virtualization technology.

Key Characteristics and Benefits

These are the primary outcomes and advantages of adopting virtualization.

1. **Hardware Independence**
   - emulated and standardized software-based copies
   - easily move to another virtual host
   - automatically resolve hardware-software incompatibility issues
   - cloning and manipulating virtual IT resources is much easier than duplicating physical hardware
2. **Server Consolidation**
   - Multiple virtual servers share one physical servers
   - Increase hardware utilization, load balancing, and optimization of available IT resources
   - Different virtual servers can run different guest operating systems on the same host.
3. **Resource Replication**
   - Virtual disk images contain binary file copies of hard disk content

- Host’s operating system can replicate, migrate, and back up the virtual server by using file operations (i.e. copy, move paste).
- Replication is one of the most salient features of virtualization technology and it enables:
  - Pre-packaging in virtual disk images in support of instantaneous deployment
  - Rapidly scale out and up
  - The ability to roll back
  - Efficient backup and restoration procedures

Types of Virtualization

1. Hardware-Based Virtualization

- Enables multiple virtual servers to interact with the same hardware platform to optimize performance.
- It overhead Issue compatibility with hardware devices and VMM (Virtual Machine Monitor).

2. Operating System-Based Virtualization

- Host operating system can provide hardware devices and rectify hardware compatibility issues,
- Hardware IT resources to be more flexibly used due to hardware independence ,
- Operating system-based virtualization can introduce demands and issues related to performance overhead (consume of Hardware IT resources, hardware-Related call, Licenses for host OS and guest OS) .
- Processing overhead required due to the running cost of host OS and virtualization software

Virtualization Management

- Modern virtualization software can automate administration tasks and reduce the overall operational burden on virtualized IT resources.
- Virtualization infrastructure management (VIM) tools support virtualized IT resource management.

Other Considerations

- Performance Overhead
- Special Hardware Compatibility
- Portability

(d) How is the hypervisor implemented in a virtualization infrastructure to generate virtual server instances from a physical server?

A hypervisor is a fundamental part of virtualization infrastructure that is used to generate virtual server instances of a physical server.

Implementation mechanisms

- One hypervisor per a physical server in general creating multiple virtual images for the given physical server.
- Responsible for creating VMs, increasing VM capacity, decreasing VM capacity or shutting VM down.
- Installed in bare-metal (meaning no OS installed) server for controlling, sharing and scheduling the usage of hardware resources (processing power, memory, I/O, etc.) which appear as dedicated resources to each virtual server.
- A single VIM (on any physical server) administrates multiple hypervisors across a physical server group.

(e) What's the difference between Cloud Computing and Grid Computing?

Cloud Computing

- is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.
- This cloud model is composed of five essential characteristics, three service models, and four deployment models.

Grid Computing

- Organize computing resources into logical pools
- Provide high performance: super virtual computer
- Distinct from clustering: much more loosely couple and distributed (heterogeneous and geographically dispersed)
- Distinct between cloud computing and grid computing:
  Grid computing based on middleware

(f) What are six essential characteristics of cloud computing?

The six essential characteristics of cloud computing are

1. **On-demand usage**
   - Gives the freedom to self-provision Cloud based IT resources.
   - Once configured, usage of the self-provisioned IT resources can be automated.
   - Enables the service-based and usage-driven features of cloud computing.
2. **Ubiquitous access**
   - Ability for a cloud service to be widely accessible.
   - To establish ubiquitous access, require support for a range of devices, transport protocols, interfaces and security technologies.
   - The cloud service architecture needs to be tailored.
3. **Multitenancy (and Resource Pooling)**
   - A software program that enables an instance of the program to serve different consumers.
   - Multitenancy models rely on the use of virtualization technologies.
   - IT resources can be dynamically assigned and reassigned, according to cloud service consumer demands.
   - Resource pooling is commonly achieved through multitenancy technology.
   - Difference between single-tenant and multitenant environments.
4. **Elasticity**
   - The automated ability of a cloud to transparently scale IT resources.
   - Elasticity is a core justification for the adoption of cloud computing .
   - Closely associated with Reduced Investment and Proportional Costs benefit.
5. **Measured Usage**
   - The ability of a cloud platform to keep track of the usage of its IT resources.
   - Measured usage is closely related to the on-demand characteristic.
6. **Resiliency**
   - Resilient computing is a form of failover
   - Redundant IT resources within the same cloud or across multiple clouds
   - Increase both the reliability and availability of their application

(g) What are the three stages of risk management in cloud security, and how do they work?

Risk Management

Assessment on potential impacts and challenges due to cloud adoption.

Cloud consumer performs risk management strategy:

1. Risk assessment
2. Risk treatment and
3. Risk control

1. Risk Assessment

- Analyze the Cloud Environment
- Identify potential vulnerabilities and shortcomings that threats can exploit
- Identify the risks according to probability of occurrence
- Evaluate the degree of impact that related to consumer utilize cloud-based IT resources

2. Risk Treatment

- Mitigation policies and plans are designed
- Mitigation actions: Some risks are eliminated, mitigated and other can be deal with via outsourcing or transferring the risk to another party (such as a cloud provider).

3. Risk Control

- Continuously monitor and manage risks
- Risk monitoring comprise survey the related events, review for determine the effectiveness of previous assessments and treatments, and identify the necessary any policy adjustment.

(h) Define load balancing in cloud computing and explain about specialized runtime workload distribution functions.

Load Balancing: A load balancer is programmed or configured with a set of performance and Quality of Service (QoS) rules and parameters.

Objectives:

- To optimize IT resource usage,
- To avoid overloads, and
- To maximizing throughput

A common approach to horizontal scaling:

- to balance a workload across two or more IT resources
- to increase performance and capacity

The load balancer mechanism is a runtime agent
Load balancers can perform a range of specialized runtime workload distribution functions that include:

Asymmetric Distribution:

Larger workloads are issued to IT resources with higher processing capacities.

Workload Prioritization workloads:

Workload Prioritization are scheduled, queued, discarded, and distributed workloads according to their priority levels.

Content-Aware Distribution:

Requests are distributed to different IT resources as dictated by the request content.